

Colony size as a species character in massive reef corals

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Abstract. In a study of seven massive, Caribbean corals, I have found major differences in reproductive behavior between species with large maximum colony sizes and species with smaller maximum colony sizes. Four species (*Diploria clivosa*, *D. strigosa*, *Montastrea cavernosa*, *Siderastrea siderea*) which are large ($> 1000 \text{ cm}^2$ in surface area) broadcast gametes during a short spawning season. Their puberty size is relatively large ($> 100 \text{ cm}^2$, except *M. cavernosa*). In contrast, two small massive species ($< 100 \text{ cm}^2$, *Favia fragum* and *S. radians*), and one medium-sized ($100\text{--}1000 \text{ cm}^2$, *Porites astreoides*) massive species, brood larvae during an extended season (year-round in Panama). The puberty size of the small species is only $2\text{--}4 \text{ cm}^2$. Given these close associations between maximum colony sizes and a number of fundamental reproductive attributes, greater attention should be given to the colony size distributions of different species of reef corals in nature, since many important life history and population characters may be inferred.

Introduction

The importance of size rather than age in determining the fate of an organism has received much attention recently (e.g., Blueweiss et al. 1979; Calder 1984; Caswell 1985; Hughes and Connell 1987; Ebenman and Persson 1988; Babcock 1991). Morphology, physiology, trophic relationships and ecological niche, among others, are all directly or indirectly affected by size (Peters 1983; Sebens 1987; Caswell 1988) and, no doubt, size has always played an important role in evolution (Newell 1949; Werner and Gilliam 1984).

Within species of scleractinian reef corals, rates of total colony mortality are inversely related to colony size (e.g., Highsmith et al. 1980; Hughes and Jackson 1980, 1985; Hughes and Connell 1987). Total growth potentially increases with increasing colony size, along with competitive and regenerative abilities (Buss 1980; Hughes

1984; Jackson and Coates 1986; Lang and Chornesky 1990). Total fecundity also generally increases as coral colonies become larger (Babcock 1984, 1991; Kojis and Quinn 1985; Szmant-Froelich 1985; but see Holloran 1986; Rinkevich and Loya 1986).

Between species, reef corals obviously vary greatly in their colony size structure and maximum colony size; but the relation between life history characters and size of coral species has not been well studied. Colony size at first reproduction seems to increase with increasing colony size of the species (data from Harriott 1983; Kojis and Quinn 1982; van Moorsel 1983; Babcock 1984). Colony size of a species has been related to the length of the brooding season in two sympatric species of *Agaricia* by van Moorsel (1983). Further, Szmant (1986) noted a relationship between size, reproductive mode and seasonality in Caribbean corals.

In this study, reproductive activity is examined as a function of colony size in six massive and one massive-encrusting species of corals which span the full size range present in shallow reefal habitats on the Caribbean coast of Panama. I use 95-percentile size (see later) as a parameter of the maximum colony size of a species and show that it is correlated with many reproductive characters in reef coral species.

Methods

Colony size

The sizes of six massive species, i.e., *Diploria strigosa*, *Favia fragum*, *Montastrea cavernosa*, *Porites astreoides*, *Siderastrea siderea*, *S. radians*, and one massive-encrusting species, *D. clivosa* common along the Caribbean coast of Panama were surveyed between July 1987 and August 1988 (Soong 1990). Transects 30–500 m long were conducted in areas where a particular species was abundant. For species with a wide depth range (0.5–7 m), transects vertical to the shore were used to cover the entire range; whereas transects parallel to the shore were used for species with relatively narrow depth distributions (0–2 m). There were 1 to 8 belt-transects for each species, depending on the habitat of each species. Sample size

Table 1. Frequency of fertile and infertile colonies in each size class of each species. Blank indicates no available colony in the size category

Species		Size class, surface area of live colony (cm ²)						
		0–4	4–15	15–60	60–250	250–1000	1000–4000	> 4000
<i>Favia fragum</i>	Fertile	24	69	16				
	Infertile	14	5	1				
<i>Siderastrea radians</i>	Fertile	32	177	202	23			
	Infertile	32	29	6	0			
<i>Porites astreoides</i>	Fertile	0	1	5	37	52	11	
	Infertile	4	11	21	10	3	0	
<i>Diploria clivosa</i>	Fertile	0	0	2	8	27	23	3
	Infertile	1	5	3	6	2	0	0
<i>Diploria strigosa</i>	Fertile			1	5	9	6	9
	Infertile			4	7	0	0	0
<i>Montastrea cavernosa</i>	Fertile		0	6	12	20	17	5
	Infertile		1	0	1	0	1	0
<i>Siderastrea siderea</i>	Fertile	0	0	0	9	26	13	22
	Infertile	3	4	7	10	1	0	1

ranged from 160 to 620 colonies for each transect. The maximum colony length and maximum width (perpendicular to the maximum length) was measured with a meter rule. The size of a colony was approximated by maximum length $\times$ maximum width. For most species, small colonies down to 1 cm were recognized, although for *F. fragum* and *S. siderea* colonies as small as 0.2 cm were counted.

Maximum colony size of a population would be underestimated from only samples; so the 95-percentile size was computed, herein called "maxima" (Mood et al. 1974). The 95-percentile size of a species was obtained from transect data after ranking all the colonies by size. When more than one transect was available for a particular species, the median value of 95-percentile sizes from different transects was used.

Reproductive modes and seasonality

A year-round sampling scheme to monitor reproductive activity in these seven species of Caribbean reef corals is detailed in Soong (1991). Reproductive mode (broadcasting versus brooding) and seasonality were noted for each species.

Puberty size

Puberty size, an estimate of colony size at maturation, was determined for each species. Puberty size is defined operationally as equal to the size of the smallest colony in the first bracket of 20 consecutive colonies (starting with the smallest, 2nd smallest, and so on) of which 90% or more were fertile (details of size distribution of sampled colonies are given in Table 1).

Results

Colony size

The colony size structure of each species was skewed to the right, i.e., with most colonies relatively small and only a few large colonies (Fig. 1). Based on the size structure, the 95-percentile size of each species was determined

(Table 2). The colony size of large species may be more than two orders of magnitude greater than that of small species.

Reproductive modes and puberty size

The small (*Favia fragum*, *Siderastrea radians*) and medium-sized (*Porites astreoides*) species brood larvae year round, whereas the four large species (*Diploria clivosa*, *D. strigosa*, *Montastrea cavernosa* and *Siderastrea siderea*) broadcast gametes in late summer (August–September) for external fertilization (for details see Soong 1991).

The large *D. strigosa*, *D. clivosa* and *S. siderea* each have a much larger puberty size (> 100 cm²) than the small *F. fragum* and *S. radians* (2–4 cm²); the medium-sized *P. astreoides* was found to have an intermediate puberty size (70 cm²) (Table 2). Puberty size of *M. cavernosa* is approximately 20 cm², although colonies smaller than 49 cm² had a very low fecundity. There is a positive correlation between puberty size and 95-percentile size in the seven species (Kendall's Tau = 0.62, $P = 0.05$). The ratio between the 95-percentile size and puberty size was less than 10 in smaller species, but was much greater in larger species (up to 37 in *D. strigosa*). The largest colony of *S. siderea* measured, for example, was 370 times that of puberty size.

Discussion

Large and small species

The differences in many aspects between large and small massive species which have been found here and in other studies are summarized in Table 3. Small species start reproducing early. They brood larvae and have relatively high rates of recruitment. Large species, on the other

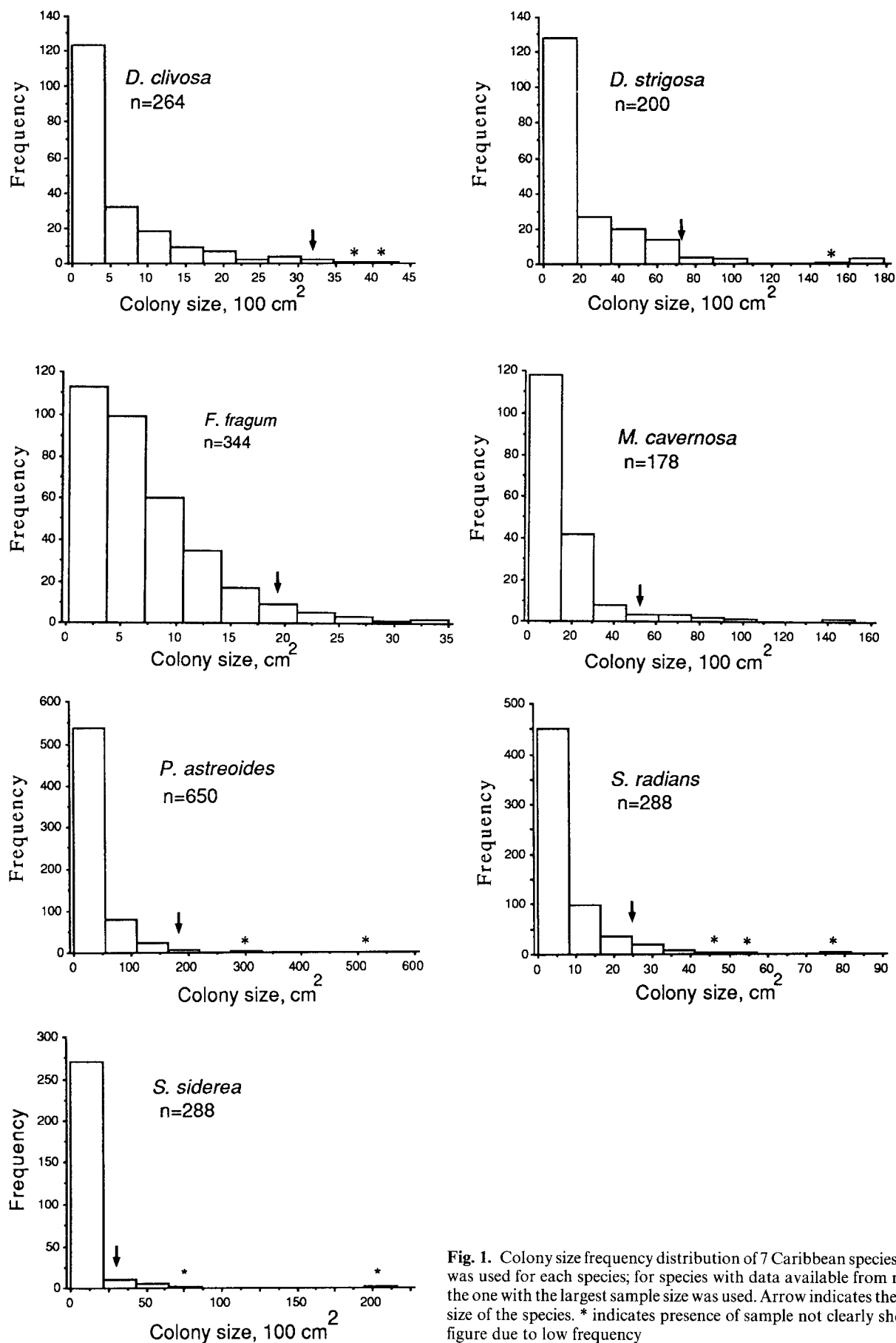


Fig. 1. Colony size frequency distribution of 7 Caribbean species. One transect was used for each species; for species with data available from many transects, the one with the largest sample size was used. Arrow indicates the 95-percentile size of the species. * indicates presence of sample not clearly shown in the figure due to low frequency

Table 2. Estimated reproductive parameters of 7 Caribbean scleractinian corals

Species	95-percentile size ^a	Reproductive mode	Puberty size (95% confidence interval) ^b	Puberty size <i>n</i>
<i>Favia fragum</i>	19 (1, 344)	Brood	2.3 (1.7–4.0)	129
<i>Siderastrea radians</i>	25 (3, 1235)	Brood	4.0 (2.3–10)	501
<i>Porites astreoides</i>	180 (8, 2841)	Brood	70 (63–165)	155
<i>Siderastrea siderea</i>	3094 (4, 936)	Broadcast	156 (110–414)	117
<i>Diploria clivosa</i>	3230 (1, 264)	Broadcast	120 (49–540)	79
<i>Montastrea cavernosa</i>	5369 (1, 178)	Broadcast	20 (12–72)	63
<i>Diploria strigosa</i>	7176 (1, 200)	Broadcast	195 (88–195)	41

^a Numbers in parenthesis indicate number of transects and number of colonies measured for estimation of 95-percentile size

^b Confidence interval estimated by a “bootstrapping” method (Diaconis and Efron 1983). The unit of all sizes in cm²

Table 3. Life history characters of massive scleractinian corals

Character	Large species	Small species	References ^a
Reproductive mode	broadcasting	brooding	2, 11, 12
Reproductive season	once, annual cycle	many times, lunar cycle	11, 12
Puberty size	large	small	13
Recruitment	low?	high?	3, 5, 6, 7, 8, 10, 11, 12
Growth rate	high	low	1, 4, 9

^a 1: Vaughan (1915); 2: Stimson (1978); 3: Bak and Engel (1979); 4: Highsmith (1979); 5: Rylaarsdam (1980); 6: Rogers et al. (1984); 7: Fitzhardinge (1985); 8: Hughes and Jackson (1985); 9: Hudson (1985); 10: Wallace (1985); 11: Szmant (1986); 12: Soong (1991); 13: this study

hand, delay reproduction until they reach larger sizes. Instead of brooding larvae, they release many gametes during annual spawning events. It is suggested that a more stable environment should be suited to large species, since continuous growth is less likely to be interrupted, whereas the short generation times of small species may be favored in unstable habitats. Hence, size in massive coral is an important species character which correlates with many reproductive traits. It is very likely that other life history characters and perhaps evolutionary rates could also be related to the size structures or maximum size of reef coral species (Potts 1984a, 1984b).

Reproductive modes and seasonality

Rinkevich and Loya (1979) suggested that the necessarily low fecundity of small polyps leads to brooding in the branching coral, *Stylophora pistillata*. Fadlallah and Pearse (1982) further proposed that, in general, mode of sexual reproduction is directly related to the skeletal design of the polyp. Nevertheless, both brooding and broadcasting are known to occur among species of *Acropora*, *Euphyllia*, *Favia*, *Pocillopora*, *Porites*, and *Siderastrea* (Table 4), despite similar polyp sizes and

skeletal characteristics within each genus (also see Szmant 1986; Harrison and Wallace 1990).

The correlation between a massive species' reproductive mode and its maximal colony size found in this

Table 4. Maximum colony size in massive genera with both reproductive modes. *S*, < 20 cm; *M*, 20–50 cm; *L*, > 50 cm in diameter

Brooding species	Broadcasting species	Size	Reference ^a
Siderastreiidae			
<i>Siderastrea radians</i>		<i>S</i>	4
	<i>S. siderea</i>	<i>L</i>	4
Poritidae			
<i>Porites astreoides</i>		<i>M</i>	4
<i>P. brighami</i>		<i>S</i>	7
<i>P. murrayensis</i>		<i>M</i>	6
<i>P. stephensoni</i>		<i>M</i>	6
	<i>P. australiensis</i>	<i>L</i>	1
	<i>P. cylindrica</i>	<i>L</i>	1
	<i>P. evermanni</i>	<i>L</i>	7
	<i>P. lobata</i>	<i>L</i>	1
	<i>P. lutea</i>	<i>L</i>	1
	<i>P. solida</i>	<i>L</i>	1
Faviidae			
<i>Favia fragum</i>		<i>S</i>	4
	<i>F. fagus</i>	<i>L</i>	6
	<i>F. lizardensis</i>	<i>L</i>	6
	<i>F. mathai</i>	<i>L</i>	6
	<i>F. pallida</i>	<i>L</i>	6
	<i>F. rotumana</i>	<i>M</i>	6
	<i>F. stelligera</i>	<i>L</i>	8
	<i>F. veroni</i>	<i>M</i>	6
<i>Cyphastrea ocellina</i>		<i>S</i>	3
	<i>C. chalcidicum</i>	<i>M</i>	5
	<i>C. microphthalma</i>	<i>M</i>	5
	<i>C. serailia</i>	<i>M</i>	5
Caryophylliidae			
<i>Euphyllia glabrescens</i>		<i>M</i>	8
	<i>E. ancora</i>	<i>L</i>	2
	<i>E. divisa</i>	<i>L</i>	2

^a Except for *Euphyllia glabrescens* (K. Soong and J. L. Chen, unpublished), mode of reproduction for each species is available in Richmond and Hunter (1990). “Size” indicates approximate maximal diameter of the species based on 1: Veron and Pichon (1982); 2: Veron (1986); 3: Wright (1986); 4: Soong (1990); 5: C. F. Dai (personal communication); 6: J. E. N. Veron (personal communication); 7: C. L. Hunter (personal communication); 8: personal observation *S*, < 20 cm; *M*, 20–50 cm; *L*, > 50 cm in diameter

study is consistent with the earlier observations of Kojis and Quinn (1982) and Szmant (1986). After examining available data, it is clear that in those genera with both modes of reproduction, massive brooders generally are smaller than massive broadcasters (Table 4). Clearly an alternative explanation for the effect of colony size on reproductive mode is required.

Rearing larvae in the coelenteron might interfere with the routine maintenance or growth of a polyp (Fadlallah and Pearse 1982). For example, planula(e) in *S. radians* often occupy and block the entrance of the coelenteron (personal observation), perhaps hindering the polyp's feeding activities (see Rinkevich and Loya 1979). Since brooding species usually carry planulae continually for a long period, especially at low latitudes (Soong 1991), the fixed cost of reproduction (energy cost which is not put into gametes but is associated with brooding, see Charnov 1979), e.g., due to feeding interference, could be high. On the other hand, broadcasting corals are not limited by these constraints, since gametes are released into the water column and embryogenesis takes place outside the body. A similar relationship between body size and reproductive mode also has been noticed in asteroids, ophiuroids, bivalves, limpets, chitons and soft corals; generally, marine invertebrates with larger body sizes broadcast gametes whereas species with smaller body size brood their young (e.g., Chia 1968; Gállardo 1977; Menge 1975); and groups showing the reverse trend are expected to be rare (Olive 1985).

Strathmann and Strathmann (1982) generalized available hypotheses correlating body size with reproductive mode into three groups, i.e., allometry, dispersal, and variable recruitment. They found that no single hypothesis explained all known cases. These three hypotheses for reef corals are discussed below.

According to the allometry hypotheses, the capacity for egg production increases faster than space for brooding as body size increases (see Strathmann and Strathmann 1982). Hence large species should opt for oviparity. However, allometric constraints appropriate for unitary organisms can hardly apply to colonial reef corals, which increase in size by adding more polyps rather than by increasing the size of individual polyps (e.g., Hughes and Jackson 1985).

In the dispersal hypotheses, only species of large body size can produce enough larvae for recruitment, which are less developed and go through prolonged planktonic stage and wide dispersal. Small species, on the other hand, must release fewer but more advanced propagules, the dispersal of which is limited. However, in the western Atlantic, species which are known to brood have a higher frequency of occurring in outlying areas (Bermuda, west Africa) than species known to be broadcasters (Jackson and Coates 1986), contrary to expectations.

According to the hypotheses of variable recruitment (Strathmann and Strathmann 1982), non-brooders have highly variable recruitment rates between years, and individuals would suffer great risk if they mature at small sizes and have short life spans. Available evidence is very sparse, but it suggests that coral recruitment is highly variable both in the Great Barrier Reefs (Wallace

1985) where broadcasting species dominate, and in the Caribbean (Hughes 1985) where 13/20 known species are brooders (Richmond and Hunter 1990). Moreover, the size distribution of coral populations indicates that natural populations of large (and necessarily long-lived), broadcasting species may have very few recruits in a particular year (Soong 1990) (This is not shown in Fig. 1 which has relatively low resolution). Hence, among the three explanations of the possible adaptive mechanisms for the correlation between reproductive mode and size of a species, the hypothesis of variable recruitment may deserve the most attention.

Puberty size

The correlation between the "maximal size" (95-percentile size in the study) of a coral species and its size of maturity (puberty size in this study) has not been reported before and suggests that, as in unitary organisms (e.g., Blueweiss et al. 1979; Calder 1984), a colonial species' size may be closely related to its life history tactics (also see Ebenman and Persson 1988). The onset of reproduction of any species should be determined by the balance between the advantages of short generation time versus postponed reproduction and the risk of increasing mortality through time (Schaffer and Gadgil 1975). Experimental and observational evidence have indicated that larger corals have lower mortality rates regardless of species (Highsmith 1982; Hughes and Connell 1987; Hughes and Jackson 1985), and that large colonies have high fecundity (e.g., Babcock 1984). For colonial organisms, their indeterminate growth, presumed freedom from senescence, high regenerative ability and their ability to spread risks of mortality spatially (due to asexual fission) could allow very high survivorship and cumulative fecundity throughout the years (e.g., Hughes and Cancino 1985). The residual reproductive value, i.e., the "sum of all future expected fecundities" (Schaffer 1974), could be very large relative to contemporary reproductive output in these organisms. A coral therefore should invest in somatic growth and delay reproduction until growth becomes limiting (Williams 1975; Hughes and Cancino 1985), assuming that present investment in reproduction will compromise further somatic growth (see Begon 1985). Nevertheless, almost every species of colonial corals studied (Harrison and Wallace 1990; Richmond and Hunter 1990) shows sexual reproduction. Apparently, despite their potentially large residual reproductive value, present engagement in reproduction is also a norm. Two hypotheses which are not necessarily mutually exclusive may be invoked to explain this enigma.

First, the intrinsic rate of skeletal growth may be constrained by colony size (Soong 1990). With increasing size, due to a nearly constant vertical or horizontal increment, the relative tissue growth rate of massive corals decreases (also see Hughes and Connell 1987), and the energy which cannot be allocated to growth could be channeled to reproduction. Second, partial mortality of corals is correlated with colony size (Hughes and Jackson 1980, 1985). Corals are adversely affected by

predation, competition and physical disturbances. Small colonies are often killed outright, large colonies may survive but sustain substantial loss of tissues. As has been shown in several field studies, large colonies are much less likely to increase in size than small surviving colonies (e.g., Connell 1973; Hughes and Jackson 1985; Hughes and Connell 1987). So with increasing colony size, even though the physiological potential for further growth remains unchanged, the expectation for size increase may be low or even negative. Implied in this second hypothesis is that after reaching a certain size, the relationship between investment in growth and increase in size is less certain (see Begon 1985), thus investment in reproduction could become favored.

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